A330 KneeBoard

VER. 25.5.22

Click for Update

by Flyingdeuk

Domestic
Japan
China
SE Asia
Oceania
Mongo, ME Asia

Differences	
FUEL Consumption	
LP Aircon/ HP StartUnit	
Cold Temp Correction	
Cold Wx Operation	
ENG ON Deicing	ENG OFF Deicing
QRH	RESET

Domestic

GMP	CJU
GMP	PUS
CJU	PUS
ICN	PUS

Welcome PA

Next Page

WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?
저는 여러분을 모시고 가는 기장 ___입니다.
저희 대한항공을 이용해 주셔서 대단히 고맙습니다.
___(국제)공항까지 비행시간은 ___시간 ___분
으로 예상됩니다.
비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,
자리에 앉아 계실 때에는 항상 좌석벨트를
매주시기 바랍니다. (저희 승무원 모두는)
저는 여러분을 안전하게 모시기 위해 최선을
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.
This is captain <u>last name</u> speaking.
Welcome aboard Korean Air.
This flight is bound for ___(international) airport and our flight time is ___ hours(s) and minutes.
For your safety, keep your seatbelts fastened while you are seated.
Thank you for choosing Koreanair.
Please enjoy your flight.

Domestic	
GMP	김포국제
ICN	인천국제
CJU	제주국제
PUS	김해국제

도착방송

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

동남아 지역 난기류 대비 기내 서비스 절차

- 6시간 미만 중거리만 1시간 전 음료서비스
- 40분전 서비스 회수
- APP PA이후 CREW 착석 강조

손님 여러분, 저는 기장입니다.

우리 비행기는 앞으로 약 (40)분 후에

___국제공항에 착륙 예정입니다.

현재 공항의 날씨는 ◐___, 기온은 섭씨 ___도 입니다.

◐ 맑으며	◐ 눈이 오고 있으며
◐ (다소)흐리며	◐ 안개가 끼어 있으며
◐ (이슬)비가 내리며/소나기가 내리며	◐ 황사가 있으며
◐ 바람이 불고 있으며	

지금 이곳의 시각은 ___월 ___일 ___요일, 오전(오후)

___시 ___분 입니다. 고맙습니다.

Ladies and gentlemen, this is the captain speaking.

We expect to land at ___ international airport in about

(40) minutes.

The current temperature at ___ is ___ degrees Celsius,

or ___ degrees Fahrenheit (NAVBLUE 참조)

and it is ◐___.

◐ (mostly) clear	◐ snowing
◐ (partly) cloudy	◐ foggy
◐ drizzling / raining	◐ hazy or smoggy
◐ windy	

The current time is ___ : ___ a.m(p.m), on (day-of-the-week), (month)(date).

Thank you for flying with us today.

Japan

GMP	HND
ICN	KIX
ICN	NRT
ICN	HND
ICN	FUK
ICN	NGO

Welcome PA

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Please enjoy your flight.

Japan	
KIX	오사카/간사이
HND	도쿄/하네다
NRT	도쿄/나리타
CTS	삿포로/신(New) 치토세
NGO	나고야/주부(Chubu Centrair)
FUK	후쿠오카

China

ICN	HKG
ICN	PEK
ICN	WUH
ICN	CAN
ICN	XIY

Welcome PA
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China	
HKG	홍콩
PEK	베이징/소우뚜(Capital)
WUH	우한/텐허
CAN	광저우/바이윈
XIY	시안/시엔양
TSN	텐진/빈하이
DLC	다렌/쪼우쑤이쯔

SE Asia

PUS	BKK
ICN	BKK
ICN	SIN
ICN	DAD
ICN	SGN
ICN	CNX

Welcome PA

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S.E Asia	
BKK	방콕/수완나폼
SIN	싱가포르/창이
DAD	다낭
SGN	베트남 호찌민/탄소넛
CNX	치앙마이
MNL	필리핀 마닐라/니노이 아키노
TPE	타이완/타이페이 타오유엔

도착 방송

Oceania

NRT	HNL
ICN	SYD
ICN	BNE

Welcome PA

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Oceania	
HNL	호놀루루/다니엘 케이 이노우에
SYD	시드니/킹스폴드 스미스
BNE	브리즈번

도착방송

Mongo, ME Asia

ICN	UBN

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Mongo, ME Asia	
UBN	울란바타르/칭기스칸

<u>도착방송</u>

RKSS(GMP) 59ft	RKPC(CJU) 119ft
KE GMP 131.15 DCL -15분 가능 TOBT5분 차이시 CTC Comm **PA**	KE CJU 129.4

Rwy 32R **Takeoff**
(06:00L~0900L / 12:00L~15:00L /18:00L~21:00L)

GMP : NADP FX-10도, CLB, CONF 1+F, Acc 4000ft

32L/R	**BULTI xT** (BULTI xQ)	**HD 324 ALT 5000 Mod NADP**
14L/R	**BULTI xU** (BULTI xZ)	**HD 144 ALT 6000 Mod NADP**

KIP 113.6	**32L IKMO 108.3**	**32R ISKP 110.7**	**14L ISEL 109.9**	**14R IOFR 108.7**

FIX	32L/R : EO32L/R, R225, YJU R271	14L/R : EO14L/R, R220, P73 / 2

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)

CJU [066/246] : STAR
AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)

ILS Z 07	**DOTOL xP**	**YUMIN**	**DOTOL 160**

RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)

ILS Z 25	**DOTOL xT(xM)**	**DUKAL**	**DOTOL/-10 160**

NAV	**YDM 25 ICHE**
FIX	RWxx /8

07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)
25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)

Entering Rapid TWY CTC GND 121.675 (STOP x)
HST 40KTS

RKPC(CJU) 119ft	RKSS(GMP) 59ft
KE CJU 129.4 DCL -10분	**PA** KE GMP 131.15

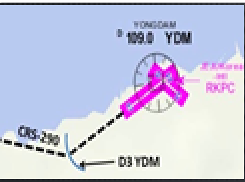
Rwy 32L **Landing**
(06:00L~0900L / 12:00L~15:00L
/18:00L~21:00L)

CJU : SID (NADP 1)

07	**KAMIT xE**	**HD 066 ALT 10000**
25	**KAMIT xW**	**HD 246 ALT 10000**

YDM 109.0	**07 109.9**	**25 ICHE 111.3**
HUD	07 : NONE	25 : YDM246/3, R290

07 : Passing G4 CTC TWR
25 : 31 Holding PSN on P, E1,2,3 CTC TWR



GMP [324/144] : STAR

ILS 32L/R	**OLMEN xT**	**BUMSI**	**OLMEN 160**
ILS 14R	**OLMEN xU**	**DOKDO**	**OLMEN 160**

NAV	**KIP 32L IKMO, 32R ISKP, 14L ISEL, 14R IOFR**
FIX	KIP /8(RWY 32), YJU R271, P73 /2

32L : D3(6532'), E2(9117'), 32R(No CL L/T) : E1(6614')
14R : C1(6578')

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G
FAF : Final Flap
TWR -> GND -> APRON (All by ATC)
Except RWY14R Landing (Until R)

RKSS(GMP) 59ft	RKPK(PUS) 13ft
KE GMP 131.15 DCL -15분 가능 TOBT5분 차이시 CTC Comm	**PA** KE Gimhae 129.2

L R
R L

Rwy 32R Takeoff
(06:00L~0900L / 12:00L~15:00L
/18:00L~21:00L)

GMP : NADP FX-10도, CLB, CONF 1+F, Acc 4000ft

32L/R	**OSPOT xT** (OSPOT xQ)	**HD 324 ALT 5000 Modi NADP**
14L/R	**OSPOT xU** (OSPOT xZ)	**HD 144 ALT 6000 Modi NADP**

KIP 113.6	**32L IKMO 108.3**	**32R ISKP 110.7**	**14L ISEL 109.9**	**14R IOFR 108.7**

HUD	32L/R : EO32L/R, R225, YJU R271	14L/R : EO14L/R, R220, P73/2

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)

PUS [002/182] : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36	**KEVOX x**	**MASTA**	9DME LG, 8DME FLAP
VOR 18	**GAYHA x**	**MASTA**	**18 Circling Click!!**
NAV	**KMH, 36L IKMA, 36R IKHE**		
FIX	36 : IKMA/IKHE /9, /8	18 : KMH R284, R280	

36L : C4 (6299'), C2(7795') 36R : E3(5866'), E2(7339')
18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only. Max Taxi SPD 20KTS
C2 HOLD SHORT 가까움(Vacate Taxi SPD)

RKPK(PUS) 13ft	RKSS(GMP) 59ft
KE Gimhae 129.2 DCL -5분	PA KE GMP 131.15

Rwy 32L Landing
(06:00L~0900L / 12:00L~15:00L
/18:00L~21:00L)

R L
L R

PUS : NADP CLB 1500, 360000lbs TOGA, CONF 1+F

36	**SOORO x** **KALOD tx**	**HD 002 ALT ATC Modi NADP**	
18	**GIMHAE x**	**HD 182 ALT 5000 Modi NADP**	
KMH 113.8	**PSN 114.0**	**36L IKMA 108.5**	**36R IKHE 109.5**

FIX	36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS

GMP [324/144] : STAR

ILS 32L/R	**GUKDO xT**	**BUMSI**	**GUKDO 160**
ILS 14R	**GUKDO xU**	**DOKDO**	**GUKDO 160**

NAV	KIP 32L IKMO, 32R ISKP, 14L ISEL, 14R IOFR
FIX	KIP /8(RWY 32), YJU R271, P73 /2

32L : D3(6532'), E2(9117'), 32R (No CL L/T) : E1(6614')
14R : C1(6578')

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G
FAF : Final Flap
TWR -> GND -> APRON (All by ATC)
Except RWY14R Landing (Until R)

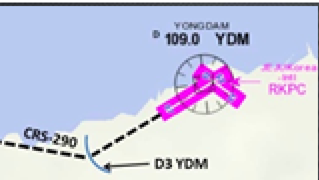
RKPC(CJU) 119ft	RKPK(PUS) 13ft
KE CJU 129.4 DCL -10분	PA KE Gimhae 129.2

CJU : SID (NADP 1)

07	AKPON xE	HD 066 ALT 9000
25	AKPON xW	HD 246 ALT ATC

YDM 109.0	07 109.9	25 ICHE 111.3
FIX	07 : NONE	25 : YDM246/3, R290

07 : Passing G4 CTC TWR
25 : 31 Holding PSN on P, E1,2,3 CTC TWR



PUS [002/182] : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36	KEVOX x	ANROD	9DME LG, 8DME FLAP
VOR 18	GAYHA x	ANROD	18 Circling Click!!
NAV	KMH, 36L IKMA, 36R IKHE		
FIX	36 : IKMA/IKHE /9, /8	18 : KMH R284, R280	

36L : C4 (6299’), C2(7795’) / 36R : E3(5866’), E2(7339’)
18R(DIS):C6(5770’),C7(6824’) 18L:E4(5882’),E5(8792’)

Vacate C3,C4 by ATC only. Max Taxi SPD 20KTS
C2 HOLD SHORT 가까움(Vacate Taxi SPD)

RKPK(PUS) 13ft	RKPC(CJU) 119ft
KE Gimhae 129.2 DCL -5분	PA KE CJU 129.4

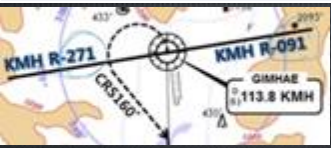
PUS : NADP CLB 1500, 360000lbs TOGA, CONF 1+F

36	SOORO x TOPAX tx	HD 002 ALT ATC Modi NADP
18	BULIM x ENGOT tx	HD 182 ALT 5000 Modi NADP

KMH 113.8	PSN 114.0	36L IKMA 108.5	36R IKHE 109.5

FIX	36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS



CJU [066/246] : STAR
AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)

ILS Z 07	UPGOS xP	YUMIN	

RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)

ILS Z 25	UPGOS xT(xM)	DUKAL	

NAV	YDM 25 ICHE
FIX	RWxx /8

07 : P6(5176’), P5(5882’), P4(6840’-ATC HIRO)
25 : P7(5219’), P8(5882’), P10(7524’-ATC HIRO)

Entering Rapid TWY CTC GND 121.675, STOP X
HST 40KTS

RKSI(ICN) 23ft	RKPK(PUS) 13ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	**PA** KE Gimhae 129.2

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**OSPOT xE/A**	**HD 333 ALT 5500/ATC SPD 230/x**
34L/R	OSPOT xY	HD 333 ALT ATC SPD 230
15L/R	**OSPOT xC**	**HD 153 ALT 5000 NADP2**
16L/R	OSPOT xH	HD 153 ALT 5000 NADP2

NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271

Parallel TWY 10KTS 이상(R17 MAX 15kts)

PUS [002/182] : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36	**KEVOX x**	**MASTA**	9DME LG, 8DME FLAP
VOR 18	**GAYHA x**	**MASTA**	**18 Circling Click!!**

NAV	**KMH KMH, 36L IKMA, 36R IKHE**	
FIX	36 : IKMA/IKHE /9, /8	18 : KMH R284, R280

36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')
18R(DIS): C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate **C3,C4** by ATC only, Max Taxi SPD 20KTS
C2 HOLD SHORT 가까움(Vacate Taxi SPD)

RKPK(PUS) 13ft	RKSI(ICN) 23ft
KE Gimhae 129.2 DCL -5분	**PA** KE ICN 131.5

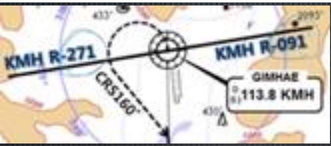
PUS : NADP CLB 1500, 360000lbs TOFA, CONF 1+F

36	**SOORO x KALOD tx**	**HD 002 ALT ATC Modi NADP**
18	**GIMHAE x**	**HD 182 ALT 5000 Modi NADP**

KMH 113.8	**PSN 114.0**	**36L IKMA 108.5**	**36R IKHE 109.5**

FIX	36 : KMH R091, R271, R185

RWY 36 400ft Man L/H turn, Max Taxi SPD 20KTS



ICN [333/153] : STAR

33/34	**GUKDO xE**	**ENPIL**	**GUKDO 180**
15/16	**GUKDO xH**	**MUNAN**	**GUKDO 180**

NAV	**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS
FIX	RWY /8(180kts), /5(160kts) , YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

RKSS(GMP) 59ft | RJTT(HND) 21ft

KE GMP 131.15 DCL -15분 가능 TOBT 5분 차이시 CTC Comm	PA	Delta Oper 132.075

Rwy 32R Takeoff
(06:00L~0900L / 12:00L~15:00L
/18:00L~21:00L)

L R
R L

GMP : NADP FX-10도, CLB, CONF 1+F, Acc 4000ft

32L/R	**EGOBA xT** (EGOBA xQ)	**HD 324 ALT 5000 Modi NADP**
14L/R	**EGOBA xU**	**HD 144 ALT 6000 Modi NADP**

KIP 113.6	**32L IKMO 108.3**	**32R ISKP 110.7**	**14L ISEL 109.9**	**14R IOFR 108.7**

HUD	32L/R : EO32L/R, R225, YJU R271	14L/R : EO14L/R, R220, P73/2

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)

DEP 125.15 – TGU 134.17 – FUK 133.02
TKO 120.5 – TKO 133.35
TKO APP 119.1 – 119.65

HND : STAR XAC Night– APP xxx Y 1400z~ SPENS 220
22/23 LDA, VOR 16L/R Tailored Procedure

34L/R	**XAC xK/H**	**KAIHO/CACAO**	**ILS X / VIS**
22[222]	**XAC 1B**	**BACON**	**LDA W(RNV22-W)**
16R/L	**XAC R**	**NATTY/SANDY**	**RNP(RNV16R/L)**
23[232]	**XAC 2B**	**DANON**	**LDA W(RNV23-W)**
NAV	**HME, 34L IHA, 34R ITC(LDA chk:22 IKL,23 ITL)**		
FIX	**22 : IKL /8, /3, HME 015** **23 : ITL/12,/7, HME 177**		**Z 34L : IHA /10, /5** **Z 34R : ITC /10, /5**

34L:L12(6515),L13(7165), 16R(DIS):L5(5147),L3(6361)
22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)
34R(DIS):C10(5780),C11(7270), 16L(DIS):C6(5925),C4(7004)

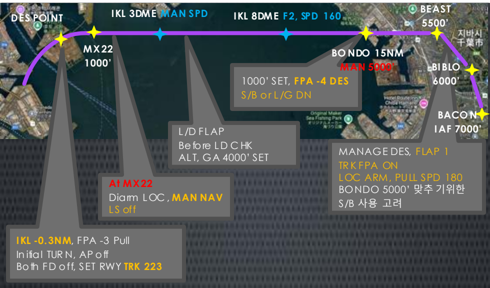
xxx Z : 180kts, 160kts limit APP Chart, xxx Y After 1400z
Gate 105P Procedure [LDA22/23 Next Page]

A330 HND LDA 22 W

F-PLN

APPR	STAR	TRANS
RNV22-W	XAC 1B	BACON

CHK IKL, IKL/8 180kts, /3 160kts, HME 015 (GA)



고려사항
- Procedure SPD 지켜야 Traffic 간격 유지 됨. 무게에 따라 -4도 증속 가능성 있음.
- S/B 사용하거나 강하각을 조절하는 것이 필요함.

Missed App

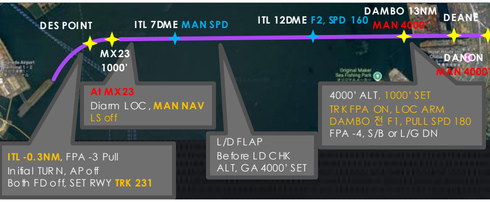
R/H Turn **HME 015** OUTBD (PULL TRK)
(DCT KASGA Radial In 196, MAN NAV)

A330 HND LDA 23 W

F-PLN

APPR	STAR	TRANS
RNV23-W	XAC 2B	BACON

CHK ITL, ITL/12 180kts, /7 160kts, HME 177 (GA)



Missed App

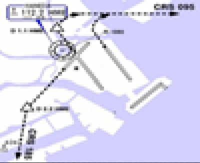
L/H Turn **HME 177** OUTBD (PULL TRK)
(DCT UTIBO Radial In 357, MAN NAV)

RJTT(HND) 21ft	RKSS(GMP) 59ft
Delta Oper 132.075 DCL -15분	PA KE GMP 131.15

Rwy 32L Landing
(06:00L~0900L / 12:00L~15:00L
/18:00L~21:00L)

HND : SID (xx B/C 2200-0230z 0600-1000z) NADP 1						
ALL	BEKLA x OPPAR x	HD 338/158,231/051,223/043 ALT ATC Mod SPD 230				
HME 112.2	34L IHA 111.7	16R ITA 111.55	34R ITC 108.9	16L IOC 111.95	22 IAD 108.1	23 ITD 110.5
FIX	34L : HME 351/1.1, R095, 34R : HME R080, R095, 22 : HME /2.2 R185					

34R BEKLA : KAIJI 230kts, TO FLAP SPD Until TORAM
05 OPPAR : Reduce SPD ARC TURN
RWY05 RTE5 TAXI Chart



DEP ATIS
TKO 120.5 – FUK 133.02
TGU 120.57
APP 119.75

GMP [324/144] : STAR			
ILS 32L/R	GUKDO xT	BUMSI	GUKDO 160
ILS 14R	GUKDO xU	DOKDO	GUKDO 160
NAV	KIP 32L IKMO, 32R ISKP, 14L ISEL, 14R IOFR		
FIX	KIP /8(RWY 32), YJU R271, P73 /2		

32L : D3(6532’), E2(9117’), 32R (No CL L/T) : E1(6614’)
14R : C1(6578’)

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G
FAF : Final Flap
TWR -> GND -> APRON (All by ATC)
Except RWY14R Landing (Until R)

RKSI(ICN) 23ft	RJBB(KIX) 17ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	KE KIX 130.95

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	EGOBA xE/A	HD 333 ALT 5500/ATC SPD 230/x		
34L/R	EGOBA xY	HD 333 ALT ATC SPD 230		
15L/R	EGOBA xC	HD 153 ALT 5000 NADP2		
16L/R	EGOBA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 134.17 – FUK 124.15

TKO 133.8

KIX RDR 120.85

KIX APP 120.25

KIX [058/238] : STAR (SAEKI 170, RANDY 150)

06L	ALISA B	BERRY	ILS Y 06L
06R	ALISA A	ALLAN	ILS Y 06R
24L/R	ALISA C	MAYAH	ILS Z 24L/R
NAV	KIE, 06L IKJ, 06R IKD, 24L IKN, 24R IKW		
FIX	RWxx /8		

06L : B8(5160'), B6(6751'), 24R : B7(5318'), B9(6751')
06R : A7(5137'), A6(6938'), 24L : A8(5269'), A9(6976')

RWY06 : After 2500ft L/G DN, After 1500ft L/D FLAP
TAXI RTE 1(via J4), 2(via J3)

RJBB(KIX) 17ft		RKSI(ICN) 23ft
KE KIX 130.95 DCL -15분	PA	KE ICN 131.5

KIX : SID – SOUJA tx (NADP 1)		
06L/R	HELEN x - SOUJA tx	HD 059 ATC (9000)
24L/R		HD 239 ATC (9000) SPD 210

KIE	06L IKJ	06R IKD	24L IKN	24R IKW
111.6	108.7	108.1	110.7	108.5

APU Start, TAXI RTE 1(via J4), 2(via J3)

DEP 119.2

TKO 132.7 – 133.8

FUK 124.15

TGU 120.57

APP 119.75

ICN [333/153] : STAR			
33/34	GUKDO xE	ENPIL	GUKDO 180
15/16	GUKDO xH	MUNAN	GUKDO 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts), YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft | RJAA(NRT) 135ft

KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA KE Tokyo 131.70

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**EGOBA xE/A**	**HD 333 ALT 5500/ATC SPD 230/x**		
34L/R	EGOBA xY	HD 333 ALT ATC SPD 230		
15L/R	**EGOBA xC**	**HD 153 ALT 5000 NADP2**		
16L/R	EGOBA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 134.17

TKO 124.15 – 132.02 – 124.1– 128.2

TKO APP 124.4 – 120.2

NRT : HAKKA 330,YAGAN 240,LIVET 210,SWAMP 150

34L/R[337]	**SWAMP E (SWAMP T)**	**ELGAR (TYLER)**	**ILS 34L/R(Z)**
16L/R[157]	**SWAMP G (SWAMP N)**	**GEMIN (NORMA)**	**ILS Z 16L/R**
NAV	**NRE, 16L ITM, 16R IKF, 34L IYQ, 34R ITJ**		
FIX	16L : ITM 4 / 34R : ITJ 14, 4 (DME) 16R : IKF 4 / 34L : IYQ 12, 4 (DME)		

16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')
16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')

L/D DOWN before 14/12 DME, L/D FLAP 4 DME
Arrival Taxi RTE in Jeppesen (No Numbering)

RJAA(NRT) 135ft	RKSI(ICN) 23ft
KE Tokyo 131.70 DCL -15분 **PA**	KE ICN 131.5

NRT : SID – ENPAR tx (NADP 1)

16L/R	TETRA x ENPAR tx CHK ALT	HD 158 ALT ATC
34L/R		HD 338 ALT 7000/ATC

NRE 117.9	16L ITM 110.7	16R IKF 111.5	34L IYQ 111.9	34R ITJ 110.9

APU Start, TAXI RTE 1, 2, 3, 4 RWY 별 DEP RTE

DEP 124.2

TKO 120.5 – 133.45 – 133.02 – 133.8

TGU 120.57

APP 119.75

ICN [333/153] : STAR

33/34	GUKDO xE	ENPIL	GUKDO 180
15/16	GUKDO xH	MUNAN	GUKDO 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	RJTT(HND) 21ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA Delta Oper 132.075

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	EGOBA xE/A	HD 333 ALT 5500/ATC SPD 230/x
34L/R	EGOBA xY	HD 333 ALT ATC SPD 230
15L/R	EGOBA xC	HD 153 ALT 5000 NADP2
16L/R	EGOBA xH	HD 153 ALT 5000 NADP2

NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 134.17 – FUK 133.02

TKO 120.5 – TKO 133.35

TKO APP 119.1 – 119.65

HND : STAR XAC Night– APP xxx Y 1400z~ SPENS 220
22/23 LDA, VOR 16L/R Tailored Procedure

34L/R	XAC xK/H	KAIHO/CACAO	ILS X / VIS
22[223]	XAC 1B	BACON	LDA W(RNV22-W)
16R/L	XAC R	NATTY/SANDY	RNP(RNV16R/L)
23[231]	XAC 2B	DANON	LDA W(RNV23-W)
NAV	HME, 34L IHA, 34R ITC(LDA chk:22 IKL,23 ITL)		
FIX	22 : IKL /8, /3, HME 015 23 : ITL/12,/7, HME 177		Z 34L : IHA /10, /5 Z 34R : ITC /10, /5

34L:L12(6515),L13(7165), 16R(DIS):L5(5147),L3(6361)
22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)
34R(DIS):C10(5780),C11(7270), 16L(DIS):C6(5925),C4(7004)

xxx Z : 180kts, 160kts limit APP Chart, xxx Y After 1400z
Gate 105P Procedure [LDA22/23 Next Page]

A330 HND LDA 22 W

F-PLN

APPR	STAR	TRANS
RNV22-W	XAC 1B	BACON

CHK IKL, IKL/8 180kts, /3 160kts, HME 015 (GA)

고려사항
- Procedure SPD 지켜야 Traffic 간격 유지 됨. 무게에 따라 -4도 증속 가능성 있음.
- S/B 사용하거나 강하각을 조절하는 것이 필요함.

Missed App

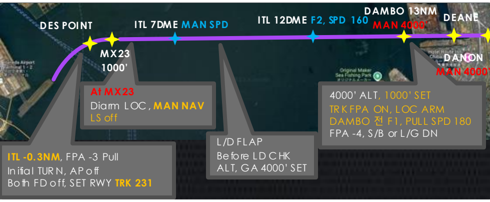
R/H Turn **HME 015** OUTBD (PULL TRK)
(DCT KASGA Radial In 196, MAN NAV)

A330 HND LDA 23 W

F-PLN

APPR	STAR	TRANS
RNV23-W	XAC 2B	BACON

CHK ITL, ITL/12 180kts, /7 160kts, HME 177 (GA)



Missed App

L/H Turn **HME 177** OUTBD (PULL TRK)
(DCT UTIBO Radial In 357, MAN NAV)

RJTT(HND) 21ft	RKSI(ICN) 23ft
Delta Oper 132.075 PA DCL -15분	KE ICN 131.5

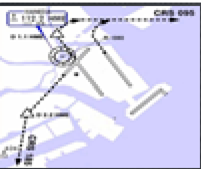
HND : SID (xx B/C 2200-0230z 0600-1000z) NADP 1

ALL	BEKLA x OPPAR x	HD 338(158), 051(231) ALT ATC Mod SPD 230

HME	34L IHA	16R ITA	34R ITC	16L IOC	22 IAD	23 ITD
112.2	111.7	111.55	108.9	111.95	108.1	110.5

FIX	34L : HME 351/1.1, R095, 34R : HME R080, R095, 22 : HME /2.2 R185

34R BEKLA : KAIJI 230kts, TO FLAP SPD Until TORAM
05 OPPAR : Reduce SPD ARC TURN
RWY05 RTE5 TAXI Chart



DEP ATIS
TKO 120.5 – FUK 133.02
TGU 120.57
APP 119.75

ICN [333/153] : STAR

33/34	GUKDO xE	ENPIL	GUKDO 180
15/16	GUKDO xH	MUNAN	GUKDO 180

NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS
FIX	RWY /8(180kts), /5(160kts) , YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

RKSI(ICN) 23ft	RJFF(FUK) 30ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA KE FUK 132.05

ICN : SID (33/34 NADP 1, 15/16 NADP 2)		
33L/R	OSPOT xE/A	HD 333 ALT 5500/ATC SPD 230/x
34L/R	OSPOT xY	HD 333 ALT ATC SPD 230
15L/R	OSPOT xC	HD 153 ALT 5000 NADP2
16L/R	OSPOT xH	HD 153 ALT 5000 NADP2

NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

TGU 125.37

Kobe 118.9 – FUK APP 119.65

FUK RDR – 121.125

FUK [338/158] : RDR Vectoring from IKE (PAVGA 13000ft) Hold W of IKE published (Confirm FMS DATA)			
16	SARUP	ENTIX	ILS, RNP, LOC 16
34	V34 HAWKS WEST	RWY34 HAWKS	VIS 34 ILS, RNP, LOC 34
NAV	DGC, 16 IFO, 34 IFF DGC VOR out of 6NM A/P		

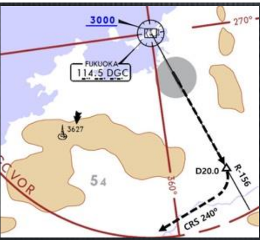
16 : C6(5505’), C7(6407’), 34 : C4(5193’), C3(6354’)

VIS 34 : After IKE – RDR Vector Downwind – 1800ft – RWY Insight 1500ft – Before L/D CHK Complete before base (Do not Extend Downwind due Terrain)

RJFF(FUK) 30ft	RKSI(ICN) 23ft
KE FUK 132.05 DCL -15min, Voice -5min	**PA** KE ICN 131.5

FUK : SID (Consider C2, C8 Intersection T/O)			
16	**HAKATA xx**	**HD 158 ALT ATC (10000)**	
34		**HD 338 ALT ATC (10000)**	
DGC 114.5		**16 IFO 111.7**	**34 IFF108.9**
FIX	16 : DGC 156/20 R240 (DGC VOR out of 6NM A/P)		

Caution GP HOLD LINE
Initial CTC TWR, “Ready for departure”
RWSL(Runway Status Lights) in operation



DEP 127.9
Kobe 135.65
TGU 125.37

ICN [333/153] : STAR			
33/34	**GUKDO xE**	**ENPIL**	**GUKDO 180**
15/16	**GUKDO xH**	**MUNAN**	**GUKDO 180**
NAV	**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

RKSI(ICN) 23ft	RJGG(NGO) 12ft
KE ICN 131.5 DCL -10분 TOBT5분 차이시 CTC Comm	PA SWISSPORT OPERATION 132.05

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**EGOBA xE/A**	**HD 333 ALT 5500/ATC SPD 230/x**		
34L/R	EGOBA xY	HD 333 ALT ATC SPD 230		
15L/R	**EGOBA xC**	**HD 153 ALT 5000 NADP2**		
16L/R	EGOBA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**
FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271		

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15

TGU 134.17 – TKO 133.8 – 133.02

센트레아 APP – 121.05

NGO [356/176] : STAR (SAMON 290, MARIA 130)

36	**CHESS(CARDS) SOUTH**	**PROBE**	**ILS Z 36**
18	**CHESS(CARDS) NORTH**	**QUEST**	**ILS Z 18**
NAV	**CBE, 36 ICX, 18 ICY**		

36 : A6(5213'), A7(6525'), A8(7837')
18 : A5(5393'), A4(6528'), A3(7841')

RWY36 : After 1500ft L/D FLAP
RWY 18 : After 3000ft L/G DN & L/D FLAP
Caution Stop line, Yellow Ramp line, VDGS!!!

RJGG(NGO) 12ft	RKSI(ICN) 23ft
SWISSPORT OPERATION 132.05 DCL -15분 **PA**	KE ICN 131.5

NGO : SID – TANGO tx (NADP 1)

36	OUMI x - TANGO tx	HD 356 ALT ATC(7000)
18		HD 176 ALT ATC(7000)

CBE 117.8	18 ICY 109.7	36 ICX 111.9

APU Start 30min, Prepare Intersection T/O

DEP 120.0

TKO 133.55 – 133.8 – TGU 120.52

APP – 119.75

ICN [333/153] : STAR

33/34	GUKDO xE	ENPIL	GUKDO 180
15/16	GUKDO xH	MUNAN	GUKDO 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	VHHH(HKG) 28ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA HAS FLT Dispatch 131.6

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	BOPTA xA	HD 333 ALT ATC		
34L/R	BOPTA xY	HD 333 ALT ATC		
15L/R	BOPTA xC	HD 153 ALT 5000 NADP2		
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

ICN 124.52(125.72) – FUK 127.5 – TPE 125.5 – 126.7
129.1 – HKG RDR 121.3 – 126.5
DEP 122.0 – Final 119.1 – 119.35

HKG [074/254] TL110 : Terminal Tx RTE + STAR Chart
ENPET FL260, RWY 25R After TOPUN - APP Mode

07L/R 07C	ABBEY xxA SIERA xxA/C	LIMES	ILS 07L/R/C
25R/L 25C	ABBEY xxB SIERA xxB/D	RIVMI	RNAV tx ILS 25R ILS 25L/C

1500 – 1659 Z RWY 07R/5L no avail, (ALL DIS TH Except 25L)

NAV	SMT, 07L IZSL, 07R IZSR, 07C IZSC 25R ITFR, 25L ITFL, 25C ITFC

07L : C7(5882'), C8(7194'),	25R : C6(5882'), C5(7211')
07R : J7(6916'), J8(7998'),	25L : J5(6916'), J4(8192')
07C : A7(5692'), A8(7208'),	25C : A6(5127'), A56673')

Confirm Chart SPD vs ATC SPD (ATC First)
Solid Line for WideBody, APU BAN off Procedure

VHHH(HKG) 28ft	RKSI(ICN) 23ft
HAS FLT Disp 131.6 DCL 20분전 5분 차이 시 CTC Comm **PA**	KE ICN 131.5

HKG : SID + Terminal Tx RTE Chart TA 9000
NADP2 : 1000 No LVR CLB - FLAP 0 - LVR CLB
3000 Manage SPD **(NADP 1/2 for 07L/R/C)**

07L/C/R	**DALOL x E/C/A** (Z/Y/X RF LEG)	**HD 074 ALT 5000 SPD 205** **NADP 1/2 (Z/Y/X 210)**
25L/C/R	**DALOL x B/D/F**	**HD 254 ALT 5000 SPD 205** **NADP 2 TA 9000**
NAV	**SMT, 07L IZSL, 07R IZSR, 07C IZSC** **25R ITFR, 25L ITFL, 25C ITFC**	
E. O	07L: IZSL074/3,R095, IZSL/9.1, 25R: ITFR254/10,R156 07C:IZSC074/2.7,R080, IZSC/6.2, 25C:ITFC254/10,R156 07R:IZSR074/6.2,R080,MAX200kt, 25L:ITFL254/6,R145	

SID – Tx RTE Chart Many SPD Restriction

HKG DEP 123.8 – RDR 118.925
TPE 129.1 – 126.7 – 123.6 – 125.5
FUK 127.5 – ICN 125.725(124.52)
ICN – 120.72 – 126.17
APP – 119.75

ICN [333/153] : STAR

33/34	**OLMEN xE**	**ENPIL**	**OLMEN 180**
15/16	**OLMEN xH**	**MUNAN**	**OLMEN 180**
NAV	**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts), YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, **HIRO**

RKSI(ICN) 23ft	ZBAA(PEK) 116ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA Air China Beijing 132.0

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	NOPIK xA	HD 333 ALT ATC		
34L/R	NOPIK xY	HD 333 ALT ATC		
15L/R	BINIL xC	HD 153 ALT 5000 NADP2		
16L/R	BINIL xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 132.8 – DLC 132.95

TAO 133.72 – 128.15 – PEK 125.6

PEK APP 120.6 – Final 119.0

PEK [001/181] : RW01/19 main (RW36L/18R)

01/36L/36R	DUMAP x Z	AA141	ILS Z 01 (Y 36L)
19/18R/18L	DUMAP x G	AA261	ILS Z 19 (Y18R)
NAV	PEK, 01 INJ, 36L IDK, 19 ISZ, 18R ILG 36R IQU, 18L IOR		

FIX : RWxx /8(180kts), /6(160kts) TMA Max 280kts

01 : Q5(5223'), Q6(7024'), 19 3.2 :Q4(5298'),Q3(7103')
36L : P6(6276'), P7(7719'), 18R : P3(6223'), P2(7552')

APU off Procedure (GND Air Cond' & GPU)
Standard TAXI RTE in Jeppesen Chart

Meter/Feet Conversion Table

Westbound ←		Eastbound →	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZBAA(PEK) 116ft	RKSI(ICN) 23ft
Air China Beijing 132.0 DCL 40분전, Voice 20분전 (COBT/STD 15분 차이 CTC Comm) **PA**	KE ICN 131.5

PEK : SID (NADP 1) RW36R/18L Intersec T/O W2, W7

36R (01)	**MUGLO x X/Y(x X)**	**HD 001 ALT 2100m(1800m)** **Maintain RWY TRK TA9850**
18L (19)	**MUGLO x G (x J)**	**HD 181 ALT DCL(1500m)** **TA 9850**

PEK 114.7	**36R IQU 111.55**	**18L IOR 109.3**	**01 INJ 108.5**	**19 ISZ 108.9**

FIX	36R : PEK 325/11, 36L : PEK 326/13, 01 : PEK 323/9 then R124

COBT from ATIS "Enroute", Bad Wx DOTRA SID

DEP 124.4
PEK APP 120.6 – PEK 125.6
DLC 123.2 – 132.95
ICN 132.8 – APP 119.75

ICN [333/153] : STAR

33/34	**REBIT x A**	**PAMBI**	**REBIT 170**
15/16	**REBIT x H**	**MUNAN**	**REBIT 170**
NAV	**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , P518 R068, R278		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

RKSI(ICN) 23ft	ZHHH(WUH)115ft
KE ICN 131.5 DCL -10분 TOBT5분 차이시 CTC Comm PA	None

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**NOPIK xA**	**HD 333 ALT ATC**		
34L/R	NOPIK xY	HD 333 ALT ATC		
15L/R	**BINIL xC**	**HD 153 ALT 5000 NADP2**		
16L/R	BINIL xH	HD 153 ALT 5000 NADP2		
NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	
Parallel TWY 10KTS 이상(R17 MAX 15kts)				

DEP 125.15 – TGU 132.8 – DLC 132.95

TAO 133.72 – 128.15 – PEK 125.6

WUH [046/226] : WUH CTL Rep HLxxxx, RDR CTC 60NM Prepare STAR CHG

04/05R/L	**PONUD 01T/05T**	**HH310/HH357**	**ILS Z**
22/23L/R	**PONUD 02T**	**AVLEN**	**ILS Z**
NAV	**WHA, 04 IHN, 05R IMU, 22 ITS, 23L IQD,** 05L IWF, 23R IUT		
FIX : RWxx /8(180kts) from intercept final 180kts			
04 : B8(5557'), B9(6578'), 22 : B5(6738'), B4(7939') 05R : F10(5475'), F11(6597'), 23L : F7(5341'), F6(6545')			
Follow Me Car TWY K/J APU off Procedure (GND Air Cond' & GPU)			

Meter/Feet Conversion Table

Westbound ←		Eastbound →	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZHHH(WUH)115ft	RKSI(ICN) 23ft
None DCL 10-30분전 Readback to CLR	**PA** KE ICN 131.5

WUH : SID (NADP 1) Readback RWY, ALT to CLR
Report Ready for P/B to CLR

04/ 05L	BIVIP 7S	HD 046 ALT 4800m or ATC(900m) TA 9850
22/ 23R	BIVIP 6S	HD 246 ALT 4800m or ATC(900m) SPD 220/230kts TA 9850

WHA 112.2	04 IHN 109.3	22 ITS 108.5	05L IWF 111.5	23R IUT 111.5

Uncommanded Turn RNP Departure

DEP 119.575

SHA 120.55 – 120.95

ICN 132.8 – APP 119.75

ICN [333/153] : STAR			
33/34	OLMEN xE	ENPIL	OLMEN 180
15/16	OLMEN xH	MUNAN	OLMEN 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	ZGGG(CAN) 50ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA China Southern 132.0

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**BOPTA xA**	**HD 333 ALT ATC**
34L/R	BOPTA xY	HD 333 ALT ATC
15L/R	**BOPTA xC**	**HD 153 ALT 5000 NADP2**
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2

NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 –

CAN [017/197] : CAN CTL, APP Freq no Chart
Short PTN AFT IAF, RW20 Windshear, Lower GA ALT

02R/L,01L/R	**OLPAB xx**	**GG611/GG612**	**ILS Z**
20L/R,19R/L		**GG511/GG513**	**ILS Z**

NAV	**POU, 02R IDM, 01L IGW, 20L IXL, 19R IKU,** 02L IBB, 01R IOO, 20R IAA, 19L IPP

Limit : RWxx /8(180kts) from final, /6(160kts)

02R:Y10(6227'),Y8(7263'), 20L:Y12(6217'),Y14(7240')
01L:N5(6453'),N4(7755'), 19R:N8(6423'),N9(7742')

STD Taxi RTE in JEPP(GREEN x), EAT 3/4 in JEPP
APU off Procedure (GND Air Cond' & GPU)

Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

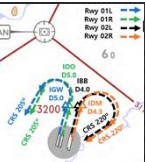
ZHHH(CAN) 50ft	RKSI(ICN) 23ft
China Southern 132.0 DCL 10-30분전 No Readback	PA KE ICN 131.5

CAN : SID (NADP 1) Extra Fuel Delay/CRZ LVL

02L/R, 01R/L	BOVMA xx RNAV	HD 017 ALT 8900m/ATC(900m) SPD 230kts TA 9850
20R/L, 19L/R		HD 246 ATC(900m) SPD 230kts TA 9850

POU 114.1 (no APT)	02R IDM 108.5	01L IGW 109.5	20L IXL 111.9	19R IKU 108.95
	02L IBB 110.35	01R IOO 109.3	20R IAA 110.75	19L IPP 111.5
FIX	01L/R : IGW/IOO D5.0 R205, 02L/R : IBB/IDM D4.0/4.3 R220			

STD Taxi RTE in JEPP(YELLOW x), Report RWY on DEP



DEP 119.7 –

ICN [333/153] : STAR

33/34	OLMEN xE	ENPIL	OLMEN 180
15/16	OLMEN xH	MUNAN	OLMEN 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	ZLXY(XIY)1572ft
KE ICN 131.5 DCL -10분 TOBT5분 차이시 CTC Comm	PA Airport Operation Center 132.0

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	NOPIK xA	HD 333 ALT ATC		
34L/R	NOPIK xY	HD 333 ALT ATC		
15L/R	BINIL xC	HD 153 ALT 5000 NADP2		
16L/R	BINIL xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 132.8 – DLC 132.95

TAO 133.725 – 128.15

PEK 125.6 – 120.35 – 133.65 – 134.15 – 126.7

XIY 125.3 – 120.95

XIY APP 119.05 – 120.2 – 125.1

XIY [052/232] : STAR TL118
Spd Restriction at REF Page
Req ILS APP instead of Visual APP

05L/R	LOVRA xx W	XY906	RNAV ILS Z 05L/R
23R/L	LOVRA xx Y	XY801	RNAV ILS Z 23R/L
NAV	LCZ, 05L IQU, 23R IRA, 05R IXW, 23L IAQ		

05L : A3(6778'), A2(9032'), 23R : A6(5544'), A7(6512')
05R : D4(5613'), D3(7322'), 23L : D5(5646'), D6(7408')

Follow Me Car, CTC Apron before Gate in
"Closing to xx TWY, apply to change to xx Freq"
Taxi RTE in Jeppesen Chart.

Meter/Feet Conversion Table

Westbound ←		Eastbound →	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZLXY(XIY) 1572ft	RKSI(ICN) 23ft
Airport Operation Center 132.0 DCL -30~10 Min, Read Back	**PA** KE ICN 131.5

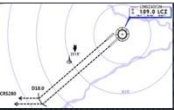
XIY (TA 9850’) : RNAV SID (NADP 1)

05L /R	WJC xx W/Z	HD 052 ALT ATC 1500m(4900’) TA 9850
23R /L	WJC xx X/Y	HD 232 ALT ATC 1500m(4900’) SPD 205/230

LCZ 109.0	05L IQU 108.55	23R IRA 111.7	05R IXW 109.3	23L IAQ 111.1

FIX	23R/L : LCZ /18

NOTAM TO Perf, ADT = CTOT



DEP 119.9– XIY 120.95 – 124.1

PEK 126.7 – 134.15 – 128.3 – 120.35

DLC 123.2 – 132.95

TAE 132.8

ICN [333/153] : STAR

33/34	REBIT x A	PAMBI	REBIT 170
15/16	REBIT x H	MUNAN	REBIT 170
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , P518 R068, R278		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

RKPK(PUS) 13ft	VTBS(BKK) 4ft
KE Gimhae 129.2 DCL -5분	PA KE Bangkok 131.25

PUS : NADP CLB 1500, 360000lbs TOGA, CONF 1+F

36	SOORO x TOPAX tx	HD 002 ALT ATC Modi NADP
18	BULIM x ENGOT tx	HD 182 ALT 5000 SPD 230 Modi NADP

KMH 113.8	PSN 114.0	36L IKMA 108.5	36R IKHE 109.5

HUD	36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS

DEP 125.5 – TGU 128.17 – 124.52(125.72)

FUK 127.5(SENKA /20)

TPE 125.5 – 129.1 – HKG 132.15 – 127.1

SNY 122.6 – HNI 123.3 – VTN 128.3

BKK 132.1 – 133.1 – APP 119.1

BKK [015/295] : TL130 UTC+7 (SPD CTL via STAR Chart)

19,20R/L	EASTE xxC	Select Tx	ILS Z 19/20L RNP 20R
01,02L/R	EASTE xxD	Select Tx	ILS Z 010/02R RNP 02L

NAV	SVB, 01 ISES, 02R ISWS, 19 ISEN, 20L ISWN
FIX	RWxx /8 (180tks), /5 (160-150kts)

19 : B8(5567'), B10(6965'), 01 : B7(5964'), B5(7962')
20L : E9(5052'), E13(7139'), 02R : E12(4872'), E7(6958')
20R : F7(6994'), F9(8471'), 02L : F6(5725'), F4(7496')

HIRO, 01/19, 02L/20R No Groov, AFT LD CONF 1
Standard Taxi Route, APU Off Procedure

VTBS(BKK) 4ft	RKPK(PUS) 13ft
KE Bangkok 131.25 DCL -20min, Voice 133.8	PA KE Gimhae 129.2

BKK : RNAV SID (NADP 1) TA 11000
A-CDM REQ Pushback +-5min of TSAT
TSAT/CTOT Inform to GND CTL

19,20L/R	LIPLI x J, G/E	HD 195 ALT 6000 19 SPD 220 TA 11000
01,02R/L	LIPLI x K, F/H	HD 015 ALT 6000 01 SPD 220 TA 11000

SVB 111.4	19 ISEN 110.5	02R ISWS 109.1	20L ISWN 109.5	01 ISES 110.1

APU Start within 10min, Standard TAXI Route
19R Do not Pass E1, D2
01/19, 02L/20R No Groov

DEP 119.25 (AUTO) – BKK 133.1 – VIE 128.3
HNI 123.3 – SNY 122.6 – HKG 127.1 – 125.35
TPE 129.1(126.7, 127.9) – 125.5
FUK 127.5 (SENKA /20)
ICN 125.725(124.52) – 128.17
APP 125.5

PUS [002/182] T/W 10tks, 36R, 360000lbs 제한

ILS 36	KEVOX x	MASTA	9DME LG, 8DME FLAP
VOR 18	GAYHA x	MASTA	18 Circling Click!!
NAV	KMH KMH, 36L IKMA, 36R IKHE		
FIX	36 : IKMA/IKHE /9, /8	18 : KMH R284, R280	

36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')
18R(DIS): C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only, Max Taxi SPD 20KTS
C2 HOLD SHORT 가까움(Vacate Taxi SPD)

RKSI(ICN) 23ft	VTBS(BKK) 4ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA
	KE Bangkok 131.25

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	BOPTA xA	HD 333 ALT ATC		
34L/R	BOPTA xY	HD 333 ALT ATC		
15L/R	BOPTA xC	HD 153 ALT 5000 NADP2		
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

FUK 127.5(SENKA /20)
TPE 125.5 – 129.1 – HKG 132.15 – 127.1
SNY 122.6 – HNI 123.3 – VTN 128.3
BKK 132.1 – 133.1 – APP 119.1

BKK [015/295] : TL130 UTC+7 (SPD CTL via STAR Chart)

19,20R/L	EASTE xxC	Select Tx (No Tx)	ILS Z 19/20L RNP 20R
01,02L/R	EASTE xxD	Select Tx (No Tx)	ILS Z 010/02R RNP 02L
NAV	SVB, 01 ISES, 02R ISWS, 19 ISEN, 20L ISWN		
FIX	RWxx /8 (180tks), /5 (160-150kts)		

19 : B8(5567'), B10(6965'), 01 : B7(5964'), B5(7962')
20L : E9(5052'), E13(7139'), 02R : E12(4872'), E7(6958')
20R : F7(6994'), F9(8471'), 02L : F6(5725'), F4(7496')

HIRO, 01/19, 02L/20R No Groov, AFT LD CONF 1
Standard Taxi Route, APU Off Procedure

VTBS(BKK) 4ft	RKSI(ICN) 23ft
KE Bangkok 131.25 DCL -20min, Voice 133.8	PA KE ICN 131.5

BKK : RNAV SID (NADP 1) TA 11000

A-CDM REQ Pushback +-5min of TSAT

TSAT/CTOT Inform to GND CTL

19,20L/R	LIPLI x J, G/E	HD 195 ALT 6000 19 SPD 220 TA 11000
01,02R/L	LIPLI x K, F/H	HD 015 ALT 6000 01 SPD 220 TA 11000

SVB 111.4	19 ISEN 110.5	02R ISWS 109.1	20L ISWN 109.5	01 ISES 110.1

APU Start within 10min, Standard TAXI Route

19R Do not Pass E1, D2

01/19, 02L/20R No Groov

DEP 119.25 (AUTO) – BKK 133.1 – VIE 128.3

HNI 123.3 – SNY 122.6 – HKG 127.1 – 125.35

TPE 129.1(126.7, 127.9) – 125.5

FUK 127.5 (SENKA /20)

ICN 125.725(124.52) – 120.72 – 126.17

ICN [333/153] : STAR

33/34	OLMEN xE	ENPIL	OLMEN 180
15/16	OLMEN xH	MUNAN	OLMEN 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

RKSI(ICN) 23ft	WSSS(SIN) 22ft
KE ICN 131.5 DCL -10분 TOBT5분 차이시 CTC Comm	PA SATS Flt Oper 131.22

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**BOPTA xA**	**HD 333 ALT ATC**
34L/R	BOPTA xY	HD 333 ALT ATC
15L/R	**BOPTA xC**	**HD 153 ALT 5000 NADP2**
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2

NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271

Parallel TWY 10KTS 이상(R17 MAX 15kts)

FUK 127.5(SENKA /20)

TPE 125.5 – 129.1

SIN [023/203] TL130 UTC+8, CPDLC WSJC -10min FIR
SPD CTL via STAR Chart, No IAF RDR Vector

02L/C/R	**MABAL/KARTO xxA**	**No Tx**	**ILS 02L**/C/R
20R/C/L	**MABAL/KARTO xxB**	**No Tx**	**ILS 20R**/C RNP 20L

RNP 02R/20L : L/D CHK -> LUVUL/OBGIS -> Arm APPR

NAV	**VTK 02L ICW, 02C ICE, 02R ICX, 20R ICH, 20C ICC**
FIX	**RWxx /8 (180tks), /4 (150kts)**

02L : W5(6519'), W4(7916'), 20R : W6(5498'), W7(6423')
02C : T6(6781'), T5 (8467'), 20C : T7 (6420'), T8 (7910')
02R : A6(6738'), A5 (7995'), 20L : A7 (6250'), A8 (7395')

HIRO Unable ATC, Exit TWY 2 way but Same Name
VTK Out of A/P, TWY R/S Max 20kts, AFT LD CONF 1

WSSS(SIN) 22ft	RKSI(ICN) 23ft
SATS Flt Oper 131.22 DCL -20min or Voice	PA KE ICN 131.5

SIN : RNAV SID (NADP 1) TA 11000
REQ P/B +-5min of TSAT, CPDLC WSJC Before DEP

20C/R 20L	TOMAN xx CHANGI xD	HD 203 ALT 3000 SPD 230 230kts Until 4000 TA11000
02C/L/R	TOMAN xx CHANGI xC	HD 023 ALT 3000 SPD 230 230kts Until 4000 TA11000

VTK 116.5	20C ICC 109.7	20R ICH 108.9	02C ICE 108.3	02L ICW 110.9

G18-21 Standard P/B, TWY R/S Max 20kts
Cross LUSMO xxx, T/O Before xxx
RWY 20C/R : CHK Min Climb Grad by NAVBLUE

DEP 120.3 – SIN

FUK 127.5 (SENKA /20)
ICN 125.725(124.52) – 120.72 – 126.17

ICN [333/153] : STAR			
33/34	OLMEN xE	ENPIL	OLMEN 180
15/16	OLMEN xH	MUNAN	OLMEN 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	VVDN(DAD) 30ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	None

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	BOPTA xA	HD 333 ALT ATC
34L/R	BOPTA xY	HD 333 ALT ATC
15L/R	BOPTA xC	HD 153 ALT 5000 NADP2
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2

NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271

Parallel TWY 10KTS 이상(R17 MAX 15kts)

FUK 127.5(SENKA /20)

TPE 125.5 – 129.1

DAD [353/173] : STAR TL100 UTC+7, C/S xxx Heavy
SPD CTL via STAR Chart, Do Not VOR 17 CAAV Auth

35L/R	BUNTA x M/N/L/K L/K : RNAV Arrs	TAMLA	ILS V 35L/R
17R/L	ILS Y 35R -> Circle	Vector	VIS 17R/L 3.5

NAV	DAN, 35L DAD, 35R IDR
FIX	RWxx /12 (200tks), /5 (160kts)

35L : G4(6702'), G6(9803'), 17R : G2(7089'), G1(9776')
35R : E4(7198'), E6(10275'), 17L : E2(8044'), E1(10754')
NO RAPID EXIT TWY, 90 Degree TURN, AFT LD CONF 1

Follow me Car on E, E2,3,4 Narrow Over-steering
TWY D 사용 금지, 소형기 on D시만 E 사용가능
Gate IN From TWY E Only, 5m 1kts Ready for STOP

VVDN(DAD) 30ft	RKSI(ICN) 23ft
None DCL -20min or Voice	**PA** KE ICN 131.5

DAD : RNAV SID (NADP 1) TA 9000
Consider Extra Fuel Due FL

35R/L	BUNTA xG/C RNAV Deps	HD 353 ALT 8000/ATC TA9000
17L/R	BUNTA xA	HD 173 ALT ATC SPD 240 TA9000

DAN 114.4	35R IDR 111.5	35L DAD 110.5

FIX	35L/R : DAN 353/5, R060 17L/R : DAN 173/4.5, R130

Complete P/B – ENG Start, Follow me Car
35R CAT I Hold Point Before E1, 17L Hold Short 가까움
TWY D 사용 금지, 소형기 on D시만 E 사용가능

DEP 120.45 – DAD

FUK 127.5 (SENKA /20)
ICN 125.725(124.52)
– 120.72 – 126.17

ICN [333/153] : STAR

33/34	OLMEN xE	ENPIL	OLMEN 180
15/16	OLMEN xH	MUNAN	OLMEN 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529’), C5(8513’), 33L : B4(7463’), B5(8513‘)
15L : C2(7522’), C1(8536’), 15R : B3(7454‘), B2(8641‘)

34L : P7(5600’), P8(6578’), 34R : N4(6876’), N5(8507‘)
16R : P6(5597’), P5(6574’), 16L : N3(7043‘), N2(8444‘)

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	VVTS(SGN) 33ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm PA	None

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	BOPTA xA	HD 333 ALT ATC		
34L/R	BOPTA xY	HD 333 ALT ATC		
15L/R	BOPTA xC	HD 153 ALT 5000 NADP2		
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

FUK 127.5(SENKA /20) – TPE 125.5 – 127.9 – 129.1
MNL 119.3 – MNL RDO 8942(5655) – HCM 120.7
132.35 – SGN APP 125.5

SGN [070/250] : STAR TL190 UTC+7 (CPDLC : VVHM)

25R(L)	DALAP xxH	SOKAN	ILS W 25R/L
07L(R)	DALAP xxG	SAMDU	ILS W 07R, VOR 07L
NAV	TSH, 25R HCM, 07R ITS, 25L SGN		
FIX	RWxx /25 (190-200tks), /5 (150-160kts)		

25R:P4(6158’), P5(6991’), 07L : P3(6266’), P2(8907’)
07R DIS :S5(6574’,110도) S3(7598’) then V(Heavy)
25L : S7(6824’), S8(9671’) Do not Confuse TWY S/S8
After Vacate then TWY V

AFT LD CONF 1, FollowMe Car Service in Ramp
(Caution Sudden STOPBAR L/T)
Sensitie VDGS!!! (0.5m이내, 2m STOP시 바로 정지)

VVTS(SGN) 33ft	RKSI(ICN) 23ft
None -15min, DEL 121.8 By Voice	**PA** KE ICN 131.5

SGN : RNP SID (NADP 1) TA 18000'
Request RWY due to Performance, Sudden M2000ft

25L(R)	**KADUM xxD/F**	**HD 250 ALT 11000/ATC** **SPD x/240 TA 18000**
07L(R)	**KADUM xxA**	**HD 070 ALT ATC TA 18000**

TSH 116.8	**25R HCM 110.5**	**07R ITS 111.7**	**25L SGN 108.3**

Caution TSAT +- 5min, **ATC CLR, RWY CHG After TAXI**
Caution Sudden STOPBAR L/T, **Follow Car Service**
1,2,3T is Stand name, Caution S1, P1 for Lineup

APP 125.5 – HCM 120.1 – 134.05
HNI 123.3 – SNY 122.6(-5min)
HKG 132.15 – 127.1 – TPE 129.1 – 127.9
126.7 – 123.6 – FUK 127.5(SENKA /20)

ICN [333/153] : STAR

33/34	**OLMEN xE**	**ENPIL**	**OLMEN 180**
15/16	**OLMEN xH**	**MUNAN**	**OLMEN 180**
NAV	**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

RKSI(ICN) 23ft	VTCC(CNX) 1036ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA CNX Traffic 131.5 D-ATIS Avail

ICN : SID (33/34 NADP 1, 15/16 NADP 2)				
33L/R	BOPTA xA	HD 333 ALT ATC		
34L/R	BOPTA xY	HD 333 ALT ATC		
15L/R	BOPTA xC	HD 153 ALT 5000 NADP2		
16L/R	BOPTA xH	HD 153 ALT 5000 NADP2		
NCN 113.8	33L INLL 109.3	33R INRR 108.9	15L ISLL 111.9	15R ISRR 109.1
WNG 112.9	34L IRFN 109.95	34R IRKN 108.1	16L IRKS 110.35	16R IRFS 108.55
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	
Parallel TWY 10KTS 이상(R17 MAX 15kts)				

FUK 127.5(SENKA /20)

TPE 125.5 – 129.1 – HKG 132.15 – 127.1

CNX [001/181] : TL130 UTC+7, Caution TERR BKK to CNX APP 80NM, FL160 – Rep PSN, ALT, SQ (SPD 220kts 20-25 TRK Mile, 180kts IF, 150-160kts FAF)			
36	MONLO xx A	PUKAT	ILS Z 36
18	MONLO xx B	PILEX	RNP 18
NAV	CMA, 36 ICMA		
36 : E(6558', 90도), D(7483') 18 DIS : F(6853', 90도), G(9199', 90도) Narrow RWY, No TWY CL L/T, Yellow Center Line			
Difference VDGS, After LD CONF 1			

Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

VTCC(CNX) 1036ft	RKSI(ICN) 23ft
CNX Traffic 131.5 -5min, By Voice GND	PA KE ICN 131.5

CNX : RNAV SID (NADP 1) TA 11000
After Airborne L TURN DCT MONLO then R-207

36	MONLO xN	HD 001 ALT 11000 SPD 250 TA 11000
18	MONLO xS	HD 181 ALT 6000(Init CL) SPD 230 TA 11000

CMA 116.9	36 ICMA 109.9

FIX	36 : CMA /4,/7, R180	18 : CMA /7, R180

CHK CLB Gradient, RECMD FLEX Due EGT Limit

DEP 129.6 – BKK 133.1
VIE 128.3 – HNI 123.3
SNY 122.6 – HKG 127.1 – 125.35
TPE 129.1(126.7, 127.9) – 125.5
FUK 127.5 (SENKA /20)
ICN 125.725(124.52) – 120.72

ICN [333/153] : STAR

33/34	OLMEN xE	ENPIL	OLMEN 180
15/16	OLMEN xH	MUNAN	OLMEN 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RJAA(NRT) 135ft		PHNL(HNL) 13ft
KE Tokyo 131.70 DCL -15분	PA	AOS Oper 132.0 No D-ATIS

NRT : SID – NADP 1				
16L/R	GULBO x	HD 158 ALT ATC		
34L/R		HD 338 ALT 7000/ATC		
NRE 117.9	16L ITM 110.7	16R IKF 111.5	34L IYQ 111.9	34R ITJ 110.9
APU Start, TAXI RTE 1, 2, 3, 4 RWY 별 DEP RTE				

DEP 124.2

TKO 120.5 – 133.45

HNL [042/079/259] UTC-10 TL180 CPDLC RJJJ-KZAK
Before SYVAD/DANNO/CANON/THOMA CTC HNL

08L	KLANI/BOOKE/OPACA Tx Modify/ RDR Vector	ILS 08L
04R		ILS Y 04R
26L		RNP AR/LDA 26L
NAV	HNL, 08L IHNL, 04R IIUM, (26L IEPC offset)	
FIX	RWY /8	

08L : Y(6837'), H(7887'), 26L : R5(7759')
04R : END of RWY Except ATC, NO LAHSO to ATC

AFT RWY 08L L/D : TWY H to A Caution
RAMP IN West/East Guide Line by ATC

EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning
(Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)
RVSM CHK : CAPT/FO 20ft, PILOT/FE 75ft
(PFD/ISIS 60ft, ST-BY 300ft)
NAV DATA Input : EEP, ETP1, ETP2, EXP
HF SELCAL CHK : Jeppesen - ENT DATA Pacific
SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : CAPT/FO 200ft

BEFORE EEP (Entry Point, ERA 기준)

60min(Theshold) : A330 420NM
One Eng : CRZ SPD 310kts, 180min 1235NM
FIX 1 : EEP, FIX 2 : ETP1
FMS ALT A/P SET : DATA – ETP Page
EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

Review Contingency Procedure

- Drift Down 30도이상, 5NM, FL290이하, +-500ft
- Wx Dev 5NM 이상, +-300ft

EDTO Segment

Apply Actual Wx for Actual Divert

Diversion

- Enroute RNP downgrade
- Weather min ERA AP
- LAND ASAP by ECAM, QRH
- Only 1 GEN(IDG, APU, EMER) + Multi Fail
- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)
- Increased Fuel Consumption

ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET
EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM
Last ETP(Critical Point) Fuel less then PLAN – Continue by PIC

EXP (Exit Point)

PHNL(HNL) 13ft	RJAA(NRT) 135ft
AOS Oper 132.0 By Voice, No D-ATIS	PA KE Tokyo 131.70

HNL : UTC-10, TA 18000, CPDLC KZAK

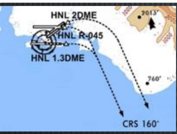
08R/L	KEOLA x Tx By Plan	HD 079 ALT 5000 TA18000
26L/R		HD 259 ALT 5000 TA18000

HNL 114.8	08L IHNL 111.7	04R IIUM 110.5

FIX	08R : HNL079/1.3, 08L : HNL R045 04L/R : HNL /2 then CRS 160

Gate Hold Proc JEPP

PushBack & Start to GND, Gate Start to RAMP



HCF 118.3

NRT [337/157] : LESPO 150

34L/R	SUPOK E (SUPOK T)	ELGAR (TYLER)	ILS 34L/R(Z)
16L/R	SUPOK G (SUPOK N)	GEMIN (NORMA)	ILS Z 16L/R

NAV	NRE, 16L ITM, 16R IKF, 34L IYQ, 34R ITJ
FIX	16L : ITM 4 / 34R : ITJ 14, 4 (DME) 16R : IKF 4 / 34L : IYQ 12, 4 (DME)

16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')
16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')

L/D DOWN before 14/12 DME, L/D FLAP 4 DME
Arrival Taxi RTE in Jeppesen (No Numbering)

RKSI(ICN) 23ft	YSSY(SYD) 21ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA Menzies SYD 129.6 No D-ATIS

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**OSPOT xE/A**	**HD 333 ALT 5500/ATC SPD 230/x**
34L/R	OSPOT xY	HD 333 ALT ATC SPD 230
15L/R	**OSPOT xC**	**HD 153 ALT 5000 NADP2**
16L/R	OSPOT xH	HD 153 ALT 5000 NADP2

NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271

Parallel TWY 10KTS 이상(R17 MAX 15kts)

TGU 125.37

Kobe 118.9 – FUK APP 119.65

FUK RDR – 121.125

SYD [335/155] UTC+10/11 TL110 RW34L/16R on Req
CHK SEMAP, Curfew 2300-0600LT, CPDLC PSN rep
ILS PRM, Noise Abate Proc Review

16L/R	**BOREE x A (x P for 16R)**	URDEN	ILS 16L/R
34R/L		SOSIJ	ILS 34L/R

NAV	SY, **34L ISN, 34R IKN, 16L ISS, 16R IKS**
FIX	RW /10(185-160kt), /5(160-150kts)

34R : U1(5515'), L(7378'), 34L : A3(6207'), A2(6620')
16LDIS:T4(5718'),T6(6971'), 16R : A3(6210'),A4(6561')
Prefer Exit Unble ATC, min REV TH, NO LASHO

SY 20NM Below 9000ft, Intl x, Dom x - TWY
Same Spot no. International/Domestic

EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning (Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)

RVSM CHK : **CAPT/FO 20ft, PILOT/FE 75ft (PFD/ISIS 60ft, ST-BY 300ft)**

NAV DATA Input : **EEP, ETP1, ETP2, EXP**

HF SELCAL CHK : Jeppesen - ENT DATA Pacific

SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : **CAPT/FO 200ft**

BEFORE EEP (Entry Point, ERA 기준)

60min(Theshold) : A330 420NM

One Eng : CRZ SPD 310kts, 180min 1235NM

FIX 1 : **EEP**, FIX 2 : **ETP1**

FMS ALT A/P SET : **DATA – ETP Page**

EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Review Contingency Procedure

- Drift Down 30도이상, 5NM, FL290이하, +-500ft
- Wx Dev 5NM 이상, +-300ft

EDTO Segment

Apply Actual Wx for Actual Divert

TERR Escape RTE Papua New Guinea

Diversion

- Enroute RNP downgrade
- Weather min ERA AP
- LAND ASAP by ECAM, QRH
- Only 1 GEN(IDG, APU, EMER) + Multi Fail
- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)
- Increased Fuel Consumption

ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET

EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Last ETP(Critical Point) Fuel less then PLAN – **Continue by PIC**

EXP (Exit Point)

YSSY(SYD) 21ft	RKSI(ICN) 23ft
Menzies SYD 129.6 **PA** No ATIS, Voice by CLR133.8/GND121.7	KE ICN 131.5

SYD : UTC+10/11, TA 10000,
CPDLC YMMM on GND 45min/Passing 10M'

16R	**GROOK x RIC tx (SYD x RDR)**	**HD 155 ALT ATC SPD 270 TA 10000**
34L	**RIC x (SYD x RDR)**	**HD 335 ALT ATC TA 10000**

SY 112.1	**34L ISN 110.1**	**16R IKS 109.5**	**34R IKN 109.3**	**16L ISS 110.9**

CASA no approval KAL (for T/O min)
CHK Special Requirement SID Chart

DEP 123.0

ICN [333/153] : STAR

33/34	**GUKDO xE**	**ENPIL**	**GUKDO 180**
15/16	**GUKDO xH**	**MUNAN**	**GUKDO 180**

NAV	**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS
FIX	RWY /8(180kts), /5(160kts) , YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

RKSI(ICN) 23ft	YBBN(BNE) 15ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA Menzies BNE 132.55

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**OSPOT xE/A**	**HD 333 ALT 5500/ATC SPD 230/x**
34L/R	OSPOT xY	HD 333 ALT ATC SPD 230
15L/R	**OSPOT xC**	**HD 153 ALT 5000 NADP2**
16L/R	OSPOT xH	HD 153 ALT 5000 NADP2

NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R,R242 YJU R271

Parallel TWY 10KTS 이상(R17 MAX 15kts)

TGU 125.37

Kobe 118.9 – FUK APP 119.65

FUK RDR – 121.125

BNE [016/196] UTC+10 TL110 Ref ALT A/P VBCG LD WT

REF SPD REST in CHART, CHK PRM Procedure

01L/R	**SMOKA x A**	**TANIK/GLENN**	ILS **01L/R**
19R/L		**OSOKO/BETSO**	ILS **19R/L**

NAV	BN, **01L IBN, 19R IBE, 01R IBA, 19L IBS**
FIX	RW xx : /5(160-150kts) into FIX Avail

01L : T6(6243'), T4(7598'), 19R : T9(6246'), T11(7595')
01R : A4(7043'), A3(8579'), 19L : A6(5620'), A7(8477')
Prefer Exit Unble ATC, min REV TH, T5, T10 No Exit

TWY B for L/D, TAXI with IDLE(Min PWR)
Short Ramp-In Lead Line, Slow Down before Turning

EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning (Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)
RVSM CHK : **CAPT/FO 20ft, PILOT/FE 75ft (PFD/ISIS 60ft, ST-BY 300ft)**
NAV DATA Input : **EEP, ETP1, ETP2, EXP**
HF SELCAL CHK : Jeppesen - ENT DATA Pacific
SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : **CAPT/FO 200ft**

BEFORE EEP (Entry Point, ERA 기준)

60min(Theshold) : A330 420NM
One Eng : CRZ SPD 310kts, 180min 1235NM
FIX 1 : **EEP**, FIX 2 : **ETP1**
FMS ALT A/P SET : **DATA – ETP Page(Nearest AP)**
EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Review Contingency Procedure

- Drift Down 30도이상, 5NM, FL290이하, +-500ft
- Wx Dev 5NM 이상, +-300ft

EDTO Segment

Apply Actual Wx for Actual Divert
TERR Escape RTE Papua New Guinea

COMM (CPDLC)

RJJJ – KZAK(FIR) – AYPM(FIR) – YBBB(FIR)
(Cross GUM airspace 250NM 118.7, SQ2100)

Diversion

- Enroute RNP downgrade
- Weather min ERA AP
- LAND ASAP by ECAM, QRH
- Only 1 GEN(IDG, APU, EMER) + Multi Fail
- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)
- Increased Fuel Consumption

ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET
EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**
Last ETP(Critical Point) Fuel less then PLAN – **Continue by PIC**

EXP (Exit Point)

YBBN(BNE)15ft	RKSI(ICN) 23ft
Menzies BNE 132.55 **PA** Voice by -5min	KE ICN 131.5

BNE : UTC+10, TA 10000, CPDLC YBBB on GND 45min/Passing 10M'		
01L/R	BIXAD x (01R Diff Chart)	HD 016 ALT ATC(6000) SPD x/240 TA 10000
19R/L	BNE x RDR SID with H/D	HD 196 ALT ATC(6000) SPD 240 TA 10000

BN 113.2	01L IBN 111.5	19R IBE 109.3	01R IBA 109.5	19L IBS 110.1

PushBack Point C,D,E , TWY A for T/O
Do Not PASS B1 for 19L
"CNL SID, TRK EXT CL, xx Degree" -> Main RWY TRK

DEP 133.45 –

ICN [333/153] : STAR			
33/34	GUKDO xE	ENPIL	GUKDO 180
15/16	GUKDO xH	MUNAN	GUKDO 180
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	ZMCK(UBN) 4485ft
KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm PA	None No D-ATIS

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	**NOPIK xA**	**HD 333 ALT ATC**
34L/R	NOPIK xY	HD 333 ALT ATC
15L/R	**BINIL xC**	**HD 153 ALT 5000 NADP2**
16L/R	BINIL xH	HD 153 ALT 5000 NADP2

NCN 113.8	**33L INLL 109.3**	**33R INRR 108.9**	**15L ISLL 111.9**	**15R ISRR 109.1**
WNG 112.9	**34L IRFN 109.95**	**34R IRKN 108.1**	**16L IRKS 110.35**	**16R IRFS 108.55**
FIX	33L/R : NC05L/R, R242 YJU R271		34L/R : EO34L/R,R242 YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 132.8 – DLC 132.95

UBN [110/290] TL 148 : Hi FE,TERR/Restrict Area
CTC UBT CTL 5min Before FIR(POLHO, INTIK)
UBT CTL ask L/D Time, **Hi FE & Sink Rate due TAS**

29	**MENUG xx W**	**ZONAM**	**ILS X 29**
11	**MENUG xx V**	**CK113**	**RNP 11 (Req)**
NAV	**SER, 29 CDA**		

29 : C(5761'), B(8241'), 11 : C(5761'), D(8231')

Gate 2 A330 Must Follow Dashed Line

Meter/Feet Conversion Table

Westbound ←		Eastbound →	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZMCK(UBN) 4485ft	RKSI(ICN) 23ft
None No D-ATIS CLR by Voice to TWR 129.0	PA KE ICN 131.5

UBN : SID (NADP 1) TA 128		
29	MENUG x C/A	HD 290 ALT ATC TA 12800
11	MENUG x D/B	HD 110 ALT ATC TA 12800
SER 116.9		29 CDA 109.9
FIX	29 : SER 290/4, R343, SER R313, SER 290/10, R260 11 : SER 110/1.5, SER R313, SER 290/10, R260	

TWY F very Narrow, TWY B,D Opposite Turn Prohibit

DEP 120.0

ICN 132.8 – APP 119.75

ICN [333/153] : STAR			
33/34	REBIT x A	PAMBI	REBIT 170
15/16	REBIT x H	MUNAN	REBIT 170
NAV	NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR		
	WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS		
FIX	RWY /8(180kts), /5(160kts) , P518 R068, R278		

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

Differences (ATS A333)

A322(5) HL7524-7551 **A323(11)** HL7553-7720	APMS	EY 252 (151) EY 260 (156)	Total PAX 200 NOR CHK OFP PAX No HL 80xx PAX AUTO Enhanced TRIM AUTO
A223E(3) HL82xx	APMS -2.5	EY 188 (112)	
A323E(6) HL80xx RNP AR, OANS PAX AUTO	APMS -1.5	EY 248 (148) HL8001 RNP AR X	
ENG START		HL7xxx 43-48%	HL8xxx 52-54%

PRINCIPAL DIMENSION

WINGSPAN : ICAO E, FAA V
Weight CAT : Heavy

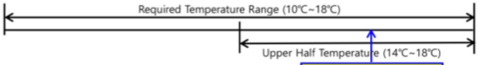
	A330-300 / 300E		A330-200 / 200E	
LENGTH	63.69 m	208.92 ft	58.37 m	191.25 ft
WING SPAN	60.30 m	197.83 ft	60.30 m	197.83 ft
HEIGHT	16.83 m	55.25 ft	17.80 m / 17.30 m	58.42 ft / 56.67 ft
WHEEL BASE	25.37 m	83.25 ft	22.18 m	72.75 ft
WHEEL TREAD	10.68 m	35.08 ft	10.68 m	35.08 ft

Refer to FCOM DSC-20-20 Principal Dimensions 180 Back 72도 48m/(157ft)

MTOW CAT ST-IN : C, Circling C/D(FAA)

238T	524,700 lbs	230T	507,063 lbs
217T	478,402 lbs	198T	436,515 lbs
192T	423,287 lbs		

LOADSHEET : CARGO No. FWD 11-21, AFT 31-41



15도 이하시 : TOD -10min 15도 SET

Middle of upper half temperature (16℃)

CRZ FUEL Penalty (Approximation)

A330
4000ft below OPT ALT : 5% increase trip fuel
8000ft below OPT ALT : 10% increase trip fuel
Above OPT ALT : 3% increase trip fuel

FUEL Consumption

APU
GND : **438 LBS/HR** (BLOCK FUEL산정)
IN FLT : **286 LBS/HR**

TAXI
2 ENG, no APU : 3300 LBS/HR **(10분 550 LBS)**

CRZ
- 100FT 당 8 MIN
 ex) **1000FT 상승 80 MIN, 1HR 20MIN**

Holding
분당 200LBS (Actual F/F 참조, 시간당 **12000LBS**)

Missed App & Landing
3000 LBS (과거 EDTO자료 1회 Missed App+L/D)

FUEL Loading

Tail Tank :

CL TANK 80000LBS 이상 5000LBS 까지

CL TANK 160000LBS 이상 10000 LBS 까지

FUEL Overfill : 1000LBS 기준

PACKS USE WITH LP AIR CONDITION UNIT (SD 연결 표시없음)

공항 요구로 APU OFF후 기내 온도 조절을 위한 방법
Low Press Unit을 연결하며 APU와 동시사용 금지

The flight crew must not use conditioned air from the packs and from the LP Air Conditioning Unit at the same time, to prevent any adverse effect on the Air Conditioning system.

절차 POM, FCOM에 없음

APU BLEED OFF 후 LP AIR 연결

EXT PWR 연결후 APU OFF

APU Start - LP Disconnect - APU BLEED ON

APU BLEED USE WITH HP AIR START UNIT (SD 연결 표시없음)

APU 부작동시 AIR START UNIT으로 시동을 위해 사용
외부 BLEED AIR의 역할을 함. APU BLEED 동시사용 금지

The flight crew must not use bleed air from the APU BLEED and from the HP Air Start Unit at the same time, to prevent any adverse effect on the Bleed Air System.

- **Before connecting the air start unit:**

PACK 1 & 2 .... as REQ F/O

HP AIR 시동전 Air Condition으로 사용시(시동시 OFF)

APU BLEED .... OFF F/O

ENG 1 & 2 BLEED .... OFF F/O

X BLEED .... OPEN F/O

AIR START UNIT CONNECT ... REQ CAP

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

ENG START
Next Page

STARTING with AIR START UNIT

통상 ENG 1 먼저 (EXT PWR Start 2 ENG)
"Req Engine Start up Present Positon~~~"

● **Before connecting the air start unit:**

PACK 1 & 2 .... OFF F/O

prevent any possible contamination of the packs by the air start unit.

APU BLEED .... OFF F/O

ENG 1 & 2 BLEED .... OFF F/O

X BLEED .... OPEN F/O

AIR START UNIT CONNECT ... REQ CAP

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

● **When cleared to start:**

Note: The AIR ABNORM BLEED CONFIG alert is triggered after the first engine start. It can be disregarded.

ENG 1 .... START CAP

Note: As necessary, engine 2 can be started first.
In this case, check the brake ACCU pressure prior to engine start.

Note: The minimum recommended starter air supply pressure is 25 PSI when the start valve is open.

● **After engine 1 is started:**

■ If the air start unit is used to start engine 2:

ENG 2 .... START CAP

● When engine 2 is started:

AIR START UNIT DISC . . . . REQ CAP

X BLEED .... AUTO F/O

ENG 1 & 2 BLEED .... ON F/O

PACK 1 & 2 .... ON F/O

■ If the crossbleed engine start procedure is used to start engine 2:

AIR START UNIT (EXT PWR) DISC.RQ CAP

ENG 1 BLEED .... ON F/O

PACK 1 & 2 .... ON F/O

CRSBLD ENG START PROC.APPLY

EXT PWR .... CHK AVAIL CAP

EXT PWR DISCONNECT .... REQ CAP

NOTE : EXT PWR removed after 1 ENG start.
but FCOM : In order to ensure that the entire network remains electrically supplied, it is recommended to start both engines before the disconnection of the external electrical power.

NOR PROC - AFTER START ....RESUME

ENG CROSSBLEED START

#1 ENG BLEED 로 #2 ENG START
Req PushBack

PushBack 완료(Parking Brake set)
One engine must be running in order to supply air for other engine start.

● Before second engine start :

APU BLEED .... OFF F/O
The BLEED valve of the supplying engine reopens and the cross bleed valve closes.
ENG BLEED (supply engine) .... ON F/O
ENG BLEED (receive engine) ..... OFF F/O
The bleed valves of receiving engines are closed to avoid reverse flow leakage.
X BLEED..................................... OPEN F/O

● When cleared to start :

AREA CLEAR OF OBS .... CONF CAP
Ensure increased power jet wake does not constitute any hazard to people or installations behind the aircraft.
THR LVR (sup eng) . ADJ BLD PRES CAP
Adjust thrust of supplying engine to obtain an engine bleed pressure of 30 PSI. If the thrust required to obtain the appropriate engine bleed pressure exceeds 40 % N1, pay particular attention to the surrounding area.
RECEIVING ENGINE .... START ALL
Apply the normal engine start procedure.

● After start :

THR LVR (supply engine) .... IDLE CAP
X BLEED..................................... AUTO F/O
ENG BLEED (receive engine) ...... ON F/O
PACK 1 &2 .... ON F/O

COLD TEMP CORRECTION General

5도 간격은 보수적으로 보간법 적용 됨

Min 제외한 모든 고도 수정은 ATC 인가 필요
Mandatory, Missed App 고도 ATC 사전 인가 없이 금지

반드시 고도 – FE 후의 고도를 보정해야함.

Ex) FE 200ft 공항 : 5000ft는 4800ft만 보정해야함.

Height Above FE (Feet) 200-800ft

TEMP	200	300	400	500	600	700	800
0	20	20	30	30	40	40	50
-5	20	30	40	40	50	60	70
-10	20	30	40	50	60	70	80
-15	30	40	50	60	80	90	100
-20	30	50	60	70	90	100	120

Height Above FE (Feet) 900-5000ft

TEMP	900	1000	1500	2000	3000	4000	5000
0	50	60	90	120	170	230	280
-5	70	80	120	160	230	310	390
-10	90	100	150	200	290	390	490
-15	110	120	180	240	360	480	600
-20	130	140	210	280	420	570	710

GMP, CJU, PUS next page

COLD TEMP CORRECTION (Domestic)

Min 은 반드시 수정 (중간 고도 CORRECTION은 PIC 결정)
Missed App 고도는 ATC 협조 필요

GMP 32L (261') / 32R (262') / 14R (254')

32L/R	8000	5500	5300	4000	2800	2300	2000
0	8450	5810	5600	4230	2970	2440	2120
-5	8620	5930	5710	4310	3030	2490	2160
-10	8780	6040	5820	4390	3080	2530	2200
R14	4000	2800	1400		4000		
0	4230	2970	1490		4230		
-5	4310	3030	1520		4310		
-10	4390	3080	1540		4390		

CJU 07 (307') / 25 (296')

	4000	2900	1800	07	8000	25	6000
0	4220	3070	1900		8450		6340
-5	4300	3130	1940		8620		6460
-10	4380	3180	1970		8780		6590

PUS 36L(233'),36R(228') / 18L/R (see below)

36L/R	6000	5000	3300	2100		6000	
0	6340	5290	3490	2210		6340	
-5	6460	5390	3560	2250		6460	
-10	6580	5490	3620	2290		6580	
18L/R	6000	5000	4000	2600	1700		6000
0	6340	5290	4230	2760	1800		6340
-5	6460	5390	4310	2810	1830		6460
-10	6580	5490	4390	2860	1870		6580

COLD Wx Operation 1/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :
• visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
• ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PRELIMINARY (OAT 3°C 이하)
PROBE/WIN HEAT – ON (ENG START후 AUTO)
LOGBOOK ------------------------------- CHECK
MAX TIME TO NEXT ENG ACC ---- DETERMINE
TAXI IN TIME – NEXT ENG ACC TIME 15분 이내인지 확인

TAXI-OUT(OAT 3°C 이하)
TAXI TIME -------------------------- MONITOR
15분 초과 또는 Eng Vibrations시 절차 수행
SOP SURFACE COND & AREA -------- CHECK
ATC ---------------------------------- NOTIFY
PARKING BRAKE ----------------------- ON
the aircraft starts to move, immediately retard the thrust levers to IDLE.
THR LEVERS -------- 50% N1 (NO HOLD TIME)
THR LEVERS --------------- RETARD TO IDLE
Repeat this procedure at intervals not longer than 15 min, or if the engine vibrations increase.
SOP TAXI ---------------------------- RESUME

BEFORE TAKEOFF
FLAPS lever ------------- SET FOR TAKEOFF
FLAPS -------------------------- CHECK PSN
T/O CONFIG pb ---------------------- TEST
T/O MEMO ----------------- CHECK NO BLUE
AFTER START checklist --------- COMPLETE
BEFORE TAKEOFF checklist -– ACCOMPLISH

TAKEOFF(OAT 3°C 이하)
A final engine acceleration must be performed before takeoff to entirely clean the engine from ice accretion.
ATC ---------------------------------- NOTIFY
PARKING BRK(OR BRK PEDALS) ----- APPLY
THR LEVERS -------- 50% N1 (NO HOLD TIME)
THR LEVERS --------------- RETARD TO IDLE
SOP TAKEOFF ----------------------- RESUME

AFTER L/D
DO NOT RETRACT FLAPS
Ice Shedding 절차 반복

COLD Wx Operation 2/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :
- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PARKING

After engine shutdown and before shutting down electrical supply:

FLAPS / SLATS .... CONF FREE OF CONTAM

Perform a visual inspection to determine if the flaps/slats mechanism is free of contamination.
If necessary, perform decontamination.

Y ELEC PUMP pb .... AUTO F/O
G ELEC PUMP pb .... AUTO F/O
Y ELEC PUMP pb (guarded) .... ON F/O
G ELEC PUMP pb (guarded) .... ON F/O

SLATS / FLAPS .... RETRACT F/O
Monitor slats/flaps retraction on ECAM upper display
When slats and flaps are retracted:

Y ELEC PUMP pb (guarded) .... AUTO F/O
G ELEC PUMP pb (guarded) .... AUTO F/O

NORMAL PROCEDURE .... RESUME ALL

ENG ON Deicing in ICN
TOBT- 40min CTC KE ICN (사전신청, 결과확인)

ICN Deicing "Deicing Required ENG On Deicing"
ICN Apron "Req Pushback Deicing Zone xx" **SQ2000**
Pad Control Arrange Deicing Pad No. **REQ FULL BODY**
Ice Man Manage Deicing Process

Gate 시동후 (FLAP UP, FLT CTL CHK 없이)

PARKING BRAKE -------------------------- SET
Report Parking Brake SET - > Ice Man
BROADBAND XMT sw ------------------- OFF
COMMUNICATION with GND ------ ESTAB
DE/ANTI-ICING FLUID TYPE ------ CHECK
CABIN PRESS MODE SEL ---- CHK AUTO
ENG 1 & 2 BLEED ------------------------- OFF
Note : **AIR ENG 1+2 BLEED FAULT** – Disregard
APU BLEED --------------------------------- OFF
DITCHING pb ------------------------------- ON
Note : **CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT OPEN, COND LAV + GAL VENT FALUT** – Disregard
THRUST LEVERS ---------------- CHK IDLE
"A/C PREPARED FOR SPRAYING"
Report Ready for Deicing - > Ice Man

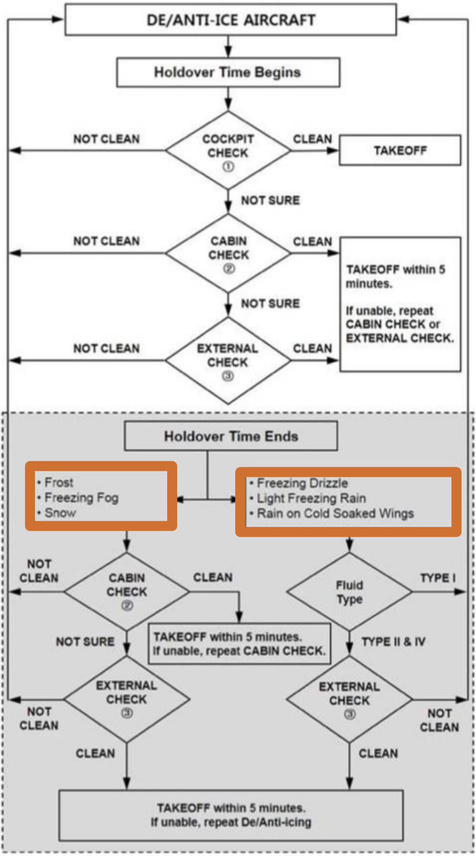
START SPRAY **REQ DCL(CTC DEL)**

UPON COMPLETE SPRAY(TIME CHK 1분)
용액과 마지막 용액 뿌린 시간 받고 적는다.
Holdover Time 결정!!!
PITOT STATICS (GND Crew) ---------- CHK
GND EQUIPMENT ------------------ REMOVE
DITCHING pb ------------------------------ OFF
OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후 **Cold Wx**
ENG 1 & 2 BLEED ---------------------------- ON
BROADBAND XMT sw --------------------- ON
5분후
APU BLEED ------------------------- AS RQRD
NORMAL PROCEDURE ----------- RESUME
FLT CONTROL CHK, FLAPS UP

DECISION TREE next page

TAKEOFF DECISION TREE



Home

ENG OFF Deicing in GMP...
TOBT- 20min CTC KE GMP (PAD, New TOBT)

REQ DCL
Deicing "Deicing Required PADxxx" ±5min TOBT
Apron "Req Pushback Deicing PADxxx"

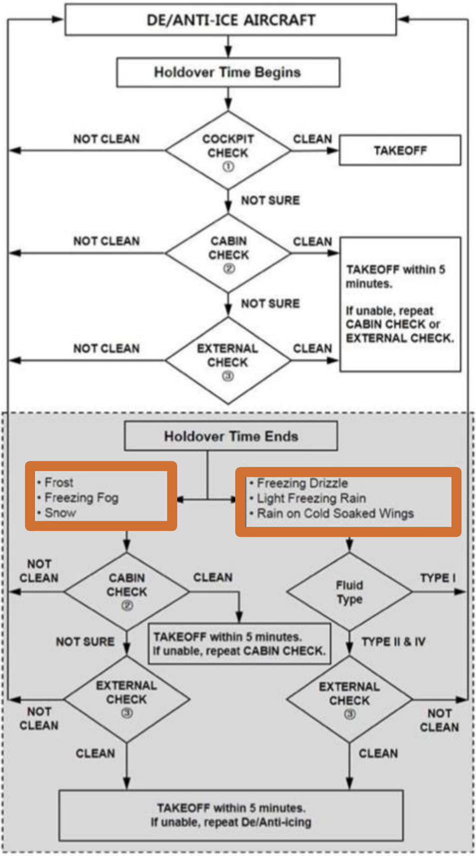
Gate 시동후 (FLAP UP, FLT CTL CHK, APU OFF 없이)
Engine anti-ice sw ------------------------- OFF
PARKING BRAKE -------------------------- SET
BROADBAND XMT sw ------------------- OFF
COMMUNICATION with GND ------ ESTAB
DE/ANTI-ICING FLUID TYPE ------ CHECK
CABIN PRESS MODE SEL ---- CHK AUTO
(HL7524-7720) GND COOL ---------- OFF
ENG 1 & 2 BLEED ------------------------ OFF
APU BLEED ---------------------------------- OFF
DITCHING pb -------------------------------- ON
Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT OPEN, COND LAV + GAL VENT FALUT –Disregard
ENG MASTER sw ------------------------ OFF
SHUTDOWN CHECKLIST
"A/C PREPARED FOR SPRAYING"

UPON COMPLETE SPRAY(TIME CHK 1분)
용액과 마지막 용액 뿌린 시간 받고 적는다.
Holdover Time 결정!!!
(HL7524-7720) GND COOL --------- OFF
PITOT STATICS (GND Crew) ---------- CHK
GND EQUIPMENT ------------------ REMOVE
DITCHING pb ------------------------------ OFF
OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후
ENG 1 & 2 BLEED --------------------------- ON
BROADBAND XMT sw -------------------- ON
5분후
APU BLEED ------------------------------------ ON
NORMAL PROCEDURE ----------- RESUME
Note : INIT B check & re-enter
PREFLT CHKlist -> Req STARTUP -> CHKlist

BOTH ENGINES ARE STARTED
FLT CTL CHK, FLAPS UP, APU as REQ
DECISION TREE next page

TAKEOFF DECISION TREE



A330 PUS VOR 18L/R

Input PERF -> SEC COPY

CLR CP18L/R, DISCONT
MDA : [ ], L/D CONF chk

FIX : KMH 280(Base Turn), 284(Missed App)

Missed App

Base Turn 이전 : L/H Turn **KMH 284** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 284** OUTBD (PULL TRK)

LOC 36 Circling
Next Page

A330 PUS LOC 36L/R Circling 18L/R

F-PLN			SEC F-PLN	
APPR	VIA	STAR	APPR	STAR
LOC36L	KEVOX	KEVOX x TRANS MASTA	18L/R	V18L/R
			(MDA: Blank)	
			(LDG CONF chk)	

CI36L(CF36R) 3500 FI36L(FF36R) 2100

FIX : KMH 280(Base Turn), 310(Missed App)

Missed App

Base Turn 이전 : L/H Turn **KMH 310** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 310** OUTBD (PULL TRK)

QRH LIST Reference Only

AIR	BLEED 1+2 FAULT(APU BLEED ON)
	BLEED 1+2 FAULT(APU BLEED NOT AVAIL)
DOOR	COCKPIT DOOR FAULT
EIS	DISPLAY UNIT FAILURE
	EIS DISPLAY DISCREPANCY
ELEC	C/B TRIPPED
	ELEC EMER CONFIG SYS-REMAINING
ENG	ALL ENG FAIL
	ENGINE FUEL SYS CONTAMINATION
	ENG RELIGHT IN FLIGHT
	ENG STALL
	ENGINE TAILPIPE FIRE
	HIGH ENGINE VIBRATION
	ON GND - NON ENG SHUTDN AFT ENG OFF
	ONE ENGINE INOPER - CIRCLING APP
	ENG1(2) EGT OVERLIMIT
F/CTL	LANDING WITH SLATS OR FLAPS JAMMED
	NO FLAPS NO SLATS LANDING
	RUDDER JAM / RUDDER PEDAL JAM
	RUDDER TRIM RUNAWAY
FUEL	FUEL IMBALANCE
	FUEL LEAK
	FUEL LOSS REDUCTION
	FUEL OVERREAD
	GRAVITY FUEL FEEDING
	TRIM TANK FUEL UNUSABLE
L/G	LANDING WITH ABNORMAL L/G
	L/G GRAVITY EXTENSION
NAV	ABNORMAL V ALPHA PROT
	ADR 1+2+3 FAULT
	UNREL SPD INDICATION
	IR ALIGNMENT IN ATT MODE
	NAV FM / GPS POS DISAGREE
	ALL ADR OFF
SMOKE	SMOKE / FUMES / AVNCS SMOKE
	REMOVAL OF SMOKE / FUMES
	SMOKE / FIRE FROM LITHIUM BATTERY
WHEEL	WHEEL TIRE DAMAGE SUSPECTED
MISC	BOMB ON BOARD
	COCKPIT WINDSHIELD/WINDOW ARCING
	COCKPIT WINDSHIELD/WINDOW CRACKED
	DITCHING
	EMER EVAC
	EMER LANDING
	FORCED LANDING
	OVERWEIGHT LANDING
	SEVERE TURBULENCE
	TAILSTRIKE VOLCANIC ASH ENCOUNTER

SYSTEM RESET TABLE 1/2

CKPT CTL pb 3초, RESET BTN 1초

A-ICE	2 WHC	GND : A.ICE L(R) WSHLD(WINDOW) HEAT
AIR	ENG BLEED pb	AIR ENG 1(2) BLEED FAULT GND : AIR ENG 1(2) BLEED NOT CLSD
	2 PACK CONT	PACK CONT RESET
COND	1 ZONE CONT	ZONE CONT RESET
APU	APU MASTER	Only APU SPD below 7%
AUTO FLT	REF FCOM	GND/Before TAXI : AUTO FLT A/THR OFF
	2 FCU	PART FCU, FCU
	2 FMGEC 2 FM	AUTO FLT FM 1(2) FAULT MAP NOT AVAIL
	MCDU OFF	MCDU RESET
BRKS	REF FCOM	GND : BRAKES RESIDUAL BRAKING WHEEL N/W STRG FAULT
CAB PR	2 CPC	FAULTY CPC RESET
COM	2 CIDS	Cabin Intercom Data Sys RESET
	2 ACP/AMU	ACP RESET
	RMP OFF	RMP RESET
	ELT RESET	ELT RESET
DATA LINK	1 ATSU	INVALID DATA or ATSU RESET
	1 AINS,1 CINS	AINS, CINS RESET
DOOR	2 PSCU	Proxi Switch Cont Unit RESET
EIS	DU OFF	ECAM, PFD, ND RESET
	3 DMC	DMC RESET
ELEC	EXT A,B OFF	GND : RESET EXT A,B
	3 BAT pb OFF	BATT RESET
	COMM, GALL pb OFF	GND : COMMER, GALL RESET
ENG	2 EIVMU	Eng Interface Vibration Monitor Unit RESET
F/CTL	PRIM,SEC OFF	GND : PRIM, SEC RESET
	2 FCDC	FCDC RESET
	2 FLAP	GND : F/CTL FLAP SYS FAULT
	2 SLAT	GND : F/CTL SLAT SYS FAULT

SYSTEM RESET TABLE 2/2

CKPT CTL pb 3초, RESET BTN 1초

FUEL	2 FCMC	GND : FUEL ZFW ZFCG DISAGREE
FWS	2 FWC, 2 SDAC	Waring Sys RESET
L/G	2 LGCIU	LGCIU RESET
MISC	VSC (Cabin aft C/B)	RESET VACUUM TOILET SYS (Blue pb)
COND	1 ZONE CONT	ZONE CONT RESET
SMOKE	2 SDCU	Smoke Dectect RESET
VENT	2 VENT CONT	RESET VENT CONT
	1 AEVC	RESET VENT COMPUTER

GS KTS	KM	MILES
300	560	350
310	570	360
320	590	370
330	610	380
340	630	390
350	650	400
360	670	410
370	690	430
380	710	440
390	720	450
400	740	460
410	760	470
420	780	480
430	800	500
440	820	510
450	830	520
460	850	530
470	870	540
480	890	550
490	910	560
500	930	580
510	950	590
520	960	600
530	980	610
540	1000	620
550	1020	630
560	1040	650
570	1060	660
580	1070	670
590	1090	680
600	1110	690
610	1130	700
620	1150	710
630	1170	730
640	1190	740
650	1200	750
660	1220	760
670	1240	770
680	1260	780
690	1280	800
700	1300	810